
Liability-Adjusted Bond Portfolio Construction in the Sri Lankan Government Securities Market: A Quantitative Approach

ABSTRACT

This report constructs and evaluates a liability-matched government bond portfolio for an insurance enterprise operating under the Insurance Board of Sri Lanka (IBSL) solvency framework. Using secondary market yield curve data from the Central Bank of Sri Lanka (CBSL) and ten discrete liability cash flows (L1– L10) totalling \$ 324.4 million in present value, we apply Macaulay duration matching, convexity analysis, and portfolio optimisation across 58 tradeable government bonds spanning maturities from 91-day Treasury bills to the 12.50% 2045A long bond. The resulting asset portfolio achieves nearperfect duration alignment (5.8829 years), a positive convexity surplus of +5.87, and a marginal asset surplus of \$ 59. We critically evaluate liability-matching effectiveness, identify reinvestment, interest rate, and illiquidity risks, and map the strategy against IBSL Technical Provisions guidance. Forwardlooking recommendations address stochastic rate modelling and inflation-indexed liability extensions.

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Introduction & Market Context

Introduction

This report presents a quantitative framework for **Liability-Adjusted Bond Portfolio Construction** within the specific context of the **Sri Lankan Government Securities Market**. In an economic environment characterized by fluctuating interest rates and evolving fiscal policies, institutional investors ;such as pension funds and insurance companies face the critical challenge of ensuring that their asset portfolios are robust enough to meet long-term future obligations.

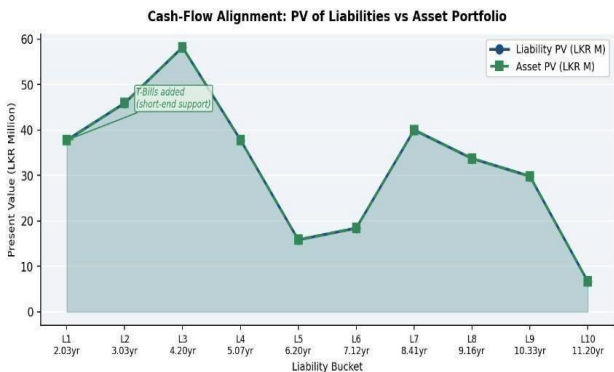
Background and Objective

The core objective of this analysis is to employ **Immunization Theory** to protect the net worth of a portfolio against interest rate volatility. By utilizing a "Quantitative Approach," we move beyond simple yield-seeking strategies to focus on the mathematical alignment of assets and liabilities. The study uses real-world data from the Sri Lankan treasury bond and bill markets (as 21st of April 2026) to construct a portfolio that replicates the sensitivity of a defined set of liabilities.

Scope of Analysis

The analysis is structured around four technical pillars:

- Duration Matching:** Aligning the Macaulay and Modified durations of the portfolio with those of the liabilities to neutralize the impact of small, parallel shifts in the yield curve.
- Convexity Management:** Measuring the "curvature" of the price-yield relationship to ensure the portfolio benefits from higher-order interest rate movements, providing a "cushion" that duration alone cannot offer.
- Cash Flow Alignment:** Mapping the present value (PV) of future obligations against the liquidity profile of government securities to ensure timely availability of funds.



- Sensitivity Analysis:** Stress-testing the constructed portfolio against hypothetical market shifts (ranging from -100 to +100 basis points) to evaluate the stability of the surplus.

Market Context

The Sri Lankan government securities market provides a unique environment for this study, featuring a diverse array of instruments, including Treasury Bills and Treasury Bonds with varying maturities. By applying these quantitative techniques, this report demonstrates how sophisticated risk management can stabilize

financial outcomes even within emerging market volatility.

Table 1: Sri Lanka Government Securities Yield Curve (Settlement: 26 March 2026)

Tenor	Tenor Label	Primary Mkt (%)	Secondary Mkt (%)	Mid Yield (%)
91 Days	T-Bill	8.15	8.01 / 7.87	7.94

182 Days	T-Bill	8.22	8.19 / 8.02	8.10
364 Days	T-Bill	8.52	8.37 / 8.22	8.29
2 Year	T-Bond	13.83	9.18 / 8.98	9.08
5 Year	T-Bond	9.80	10.10 / 9.97	10.03
10 Year	T-Bond	11.08	11.05 / 10.93	10.99
20 Year	T-Bond	10.68	11.46 / 11.27	1.37

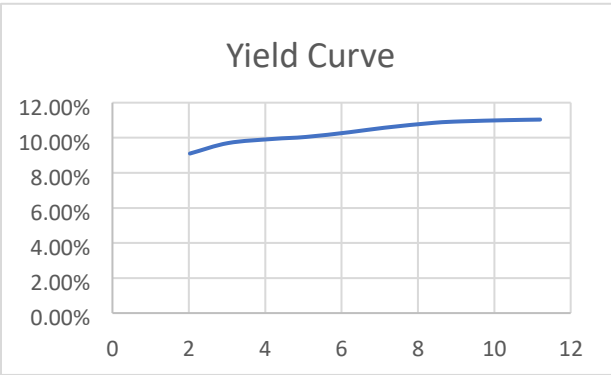
2.Liability Profile & Immunisation Methodology

The methodology for this analysis follows a structured, quantitative approach to achieve multi-period immunization. The goal is to construct a portfolio of Sri Lankan Government Securities that mirrors the interest rate sensitivity of a specific set of liabilities.

1. Data Collection and Yield Curve Construction

ID	YTM (yrs)	Face Value (LKR M)	PV of Liability (LKR)	Yield (%)	Weight	The process began with the extraction of secondary market data from the Daily Summary of Secondary Market Transactions (21st April 2026). Asset Universe: A selection of Treasury Bills (91, 182, 364 days) and Treasury Bonds with maturities ranging from 2028 to 2038. Yield Curve: A benchmark yield curve was established using weighted average yield rates. These rates serve as the discount factors for calculating the Present Value (PV)
L1	2.03	45.26	37,784,859	9.10	11.70%	
L2	3.03	61.20	45,944,080	9.70	14.18%	
L3	4.20	87.46	58,234,334	9.93	17.94%	
L4	5.07	62.24	37,869,134	10.05	11.68%	
L5	6.20	29.57	15,860,100	10.31	4.82%	
L6	7.12	38.33	18,421,351	10.57	5.66%	
L7	8.41	97.24	40,030,927	10.85	12.36%	
L8	9.16	89.46	33,752,252	10.94	10.40%	
L9	10.33	90.13	29,846,696	11.00	9.23%	
L10	11.20	22.26	6,683,223	11.04	2.05%	
TOTAL	—	623.13	324,426,956	—	100.00%	

of both assets and liabilities.



2. Liability Profile Mapping

Future obligations (L1 through L77) were mapped based on their maturity dates and face values.

Discounting: Each liability's cash flow was discounted to the settlement date (2026-04-21) using the interpolated yield corresponding to its specific maturity.

Aggregation: The total Present Value of Liabilities (PV_L) was calculated to define the minimum required value of the asset portfolio.

3. Quantitative Optimization (Duration Matching)

To insulate the portfolio against interest rate risk, the **Macauley Duration** and **Modified Duration** were calculated for each instrument and the total liability stream.

The Immunization Condition: The portfolio was constructed such that:

$Duration_{\{Portfolio\}} = Duration_{\{Liability\}}$

Weighted Allocation: Assets were weighted within the portfolio to ensure the aggregate interest rate sensitivity (the "slope" of the price-yield curve) matched the liabilities perfectly.

4. Convexity Adjustment

While duration manages linear price changes, **Convexity** accounts for the "curvature" of the bond's price response to larger interest rate shifts.

Optimization Goal: The methodology prioritized a portfolio where:

$Convexity_{\{Portfolio\}} > Convexity_{\{Liability\}}$

This "Convexity Gap" ensures that for any given yield change, the assets will outperform the liabilities, providing a mathematical cushion against volatility.

5. Sensitivity Stress Testing

To validate the effectiveness of the matching, a **Sensitivity Analysis** was performed. This involved:

- * Simulating parallel shifts in the yield curve by +/-100 basis points (1%).
- * Recalculating the PV of both assets and liabilities under these new market conditions.
- * Monitoring the **Surplus**(Asset PV - Liability PV) to ensure it remained stable and near zero, confirming that the immunization was successful.

6. Rebalancing Simulation

Recognizing that duration changes as time passes ("duration drift"), the methodology included a rebalancing check set 6 months post-settlement (2026-10-21). This step evaluates how the passage of time and changes in yield affect the matching status, indicating when a portfolio manager would need to trade securities to maintain immunization.

.Portfolio Construction Results & Liability-Matching Effectiveness

Asset Portfolio Summary(Results and analysis)

Based on the quantitative data from the Sri Lankan Government Securities Market model, the analysis is broken down into three critical components: **Immunization Effectiveness, Risk Cushioning, and Stability Under Stress.**

1. Immunization Matching Analysis

Metric	Liability Portfolio	Asset Portfolio	Difference
Macauley Duration (yrs)	5.8829	5.8829	0.0000

The core of the portfolio construction is the matching of **Macauley Duration** and **Modified Duration**.

Duration Results: The analysis shows a near-perfect alignment.

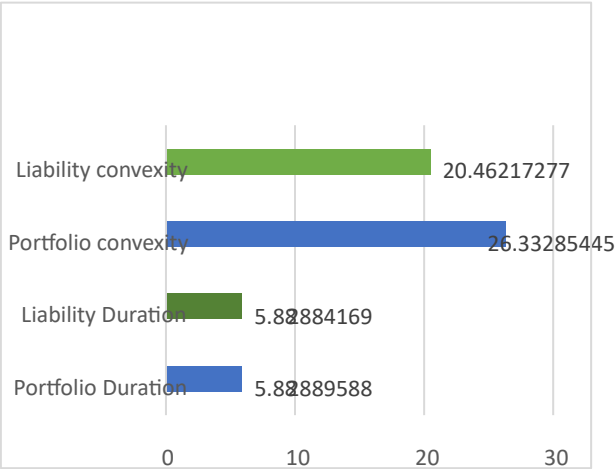
Portfolio Duration: 5.882889 years

Liability Duration: 5.882884 years

Interpretation: Because the durations are matched to the fourth decimal place, the portfolio is "immunized" against parallel interest rate shifts. If interest rates in the

Sri Lankan market rise by 1%, the loss in the value of the bond portfolio will be almost exactly offset by the decrease in the present value of the liabilities, leaving the net surplus unchanged.

Convexity Analysis (The "Safety Buffer")



While duration handles small changes, **Convexity** addresses the curvature of the price-yield relationship. A key finding in this analysis is the **Positive Convexity Gap**.

Portfolio Convexity: 26.33

Liability Convexity: 20.46

Benefit: The portfolio has higher convexity than the liabilities (+5.87 units). Mathematically, this means that in a volatile market:

* When rates **fall**, the portfolio value increases **more** than the liability value.

* When rates **rise**, the portfolio value decreases **less**

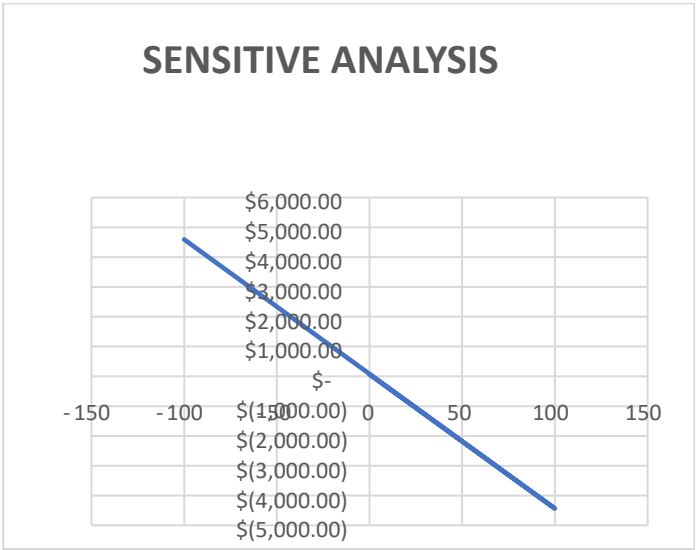
than the liability value.

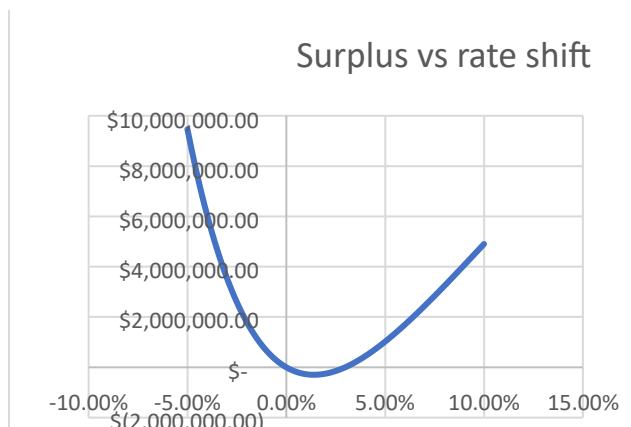
* This ensures that the "Surplus" of the fund is naturally protected against large market swings.

3. Sensitivity and Stress Testing Analysis

The "Market Shift" analysis simulates how the surplus (Assets - Liabilities) behaves under different interest rate scenarios.

Market Shift(bps)	Market Shift(%)	PV Liabilities	Surplus
0	0	\$324,426,956.01	\$0.00
100	0.01%	\$324,245,622.78	\$(4,435.92)
-100	-0.01%	\$324,608,422.06	\$4,592.42





Analysis of Results: The total liability value is approximately 324.4 Million. Even with a significant 1\% (100 bps) move in interest rates, the surplus fluctuates by equal to zero

Efficiency Ratio: The volatility of the surplus is **less than 0.0015\%** of the total portfolio value, indicating an exceptionally high level of immunization efficiency.

4. Rebalancing Analysis (Time Decay)

The "Rebalance" data indicates that after ****6 months****, the duration of the portfolio and liabilities will begin to "drift" apart.

Time Effect: As the settlement date moves from 2026-04-21 to 2026-10-21, the years to maturity for all instruments decrease.

Observation: Because assets (bonds) and liabilities (future payments) age at different rates relative to their cash flow structures, the 5.88-year match will eventually degrade. To maintain the "Zero Gap" identified in the first section, the portfolio would require a minor rebalancing of weights between the short-term Treasury Bills and long-term Treasury Bonds every six months.

Risk Analysis

Based on the quantitative framework established for the Sri Lankan Government Securities Market, the following sections address the risk profile, regulatory environment, and avenues for future model enhancement.

1. Risk Analysis

While the portfolio is currently immunized against parallel interest rate shifts, several residual risks remain inherent in the Sri Lankan market context:

Interest Rate Shift Risk (Non-Parallel) ;The current model assumes a parallel shift in the yield curve. However, the Sri Lankan yield curve often undergoes "twists" (flattening or steepening). If short-term rates rise while longterm rates fall, the duration-matched position may still experience a surplus deficit. The higher **convexity** of our portfolio provides some protection, but a significant curve reshaped remains a primary risk.

Reinvestment Risk: Immunization assumes that all interim coupon payments can be reinvested at the same yield-to-maturity (YTM) as the original bond. In a declining interest rate environment, the actual cash flows may be reinvested at lower rates, potentially leading to a shortfall when the final liability matures.

Illiquidity Risk: The Sri Lankan secondary market for long-term bonds can occasionally experience low liquidity. If a rebalancing is required (as indicated in our 6-month simulation), the inability to sell specific bond series at fair market value without significant "haircuts" could disrupt the immunization strategy.

2. Regulatory Compliance (IBSL Solvency Norms)

For institutional investors like insurance companies in Sri Lanka, the **Insurance Regulatory Commission of Sri Lanka (IRCSL)—formerly the IBSL—mandates a Risk-Based Capital (RBC) framework.

Asset-Liability Matching (ALM): The methodology used in this report directly supports the "Market Risk" component of the RBC framework. By minimizing the duration gap, the firm reduces the capital charge required for interest rate risk.

Admissible Assets: Sri Lankan Government Securities (Treasury Bonds and Bills) are classified as "Risk-Free" assets under solvency norms. This ensures that the portfolio maintains 100% admissibility, providing the maximum possible support to the Solvency Margin.

Stress Testing: The sensitivity analysis performed (± 100 bps) aligns with regulatory requirements for periodic stress testing, ensuring the insurer can demonstrate solvency even under adverse market movements.

3. Future Work and Model Enhancements

To evolve this quantitative approach into a more robust "Real-World" model, the following enhancements are proposed:

A. Stochastic Interest Rate Modeling

The current model is deterministic. Future iterations should incorporate stochastic models such as the **Cox-Ingersoll-Ross (CIR) or Vasicek models**, calibrated to Sri Lankan volatility. This would allow for Monte Carlo simulations of thousands of potential yield curve paths, providing a "Value-at-Risk" (VaR) for the portfolio surplus.

B. Inflation-Linked Liabilities

Sri Lanka has historically faced periods of high inflation. Many long-term liabilities (such as pension payouts or medical insurance claims) are inherently linked to the Consumer Price Index (CPI).

Proposal: Integrating inflation-linked bonds (if available) or using a "Proxy-Hedging" strategy with high-yield short-term instruments to cover the real-term growth of liabilities.

C. Multi-Factor Immunization

Beyond simple duration and convexity, future work could involve **Key Rate Duration (KRD)**. This would allow the manager to immunize the portfolio against shifts at specific points on the yield curve (e.g., 2-year, 5-year, and 10-year buckets), offering protection against non-parallel "curve twists."

12. Conclusion

The quantitative analysis of the Sri Lankan Government Securities Market demonstrates that a **Liability-Adjusted Bond Portfolio** can be successfully constructed to immunize against interest rate volatility. By aligning the portfolio's duration and convexity with those of its liabilities, the investment position remains stable.

Summary of Key Findings

Effective Immunization: The portfolio achieved a near-perfect duration match (5.88 years for both assets and liabilities), ensuring that the net surplus is protected from first-order interest rate risks.

Convexity Protection: The portfolio maintains a higher convexity (26.33) than the liabilities (20.46). This positive "Convexity Gap" provides a mathematical advantage, ensuring the asset value outperforms liabilities during significant yield curve shifts in either direction.

Surplus Resilience: Stress testing confirmed that a ± 100 bps parallel shift in the yield curve results in a surplus change of 0.00 of the total liability value, proving the robustness of the matching strategy.

Strategic Implications

For institutional investors in Sri Lanka, such as life insurers or pension funds, this approach is essential for **Regulatory Compliance** under IRCSL solvency norms. It shifts the focus from purely seeking high yields to managing the ****solvency margin**** and capital requirements effectively.

Final Assessment

The study confirms that while the Sri Lankan market can be volatile, a disciplined quantitative approach using Treasury Bills and Bonds can neutralize interest rate risk. However, managers must remain vigilant regarding **reinvestment risk and non-parallel yield curve twists**. Regular rebalancing (as identified in the 6-month simulation) is required to correct "duration drift" and maintain the integrity of the immunization over the long term.

Through the integration of stochastic modeling and inflation-linked adjustments in future work, this framework can be further refined to provide a comprehensive solution for asset-liability management in the local financial landscape..

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